

- ❖ **Boyle's law:** If m and T are constant

$$V \propto \frac{1}{P}$$

$$P_1 V_1 = P_2 V_2$$

- ❖ **Charles's law:** If m and P are constant

$$V \propto T$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- ❖ **Gay-Lussac's law:** If m and V are constant

$$P \propto T$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- ❖ **Avogadro's law:** If P , V and T are same

$$N_1 = N_2$$

where, N_1 and N_2 are the number of molecules

$V \propto n$ (no. of molecules of gas)

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- ❖ **Graham's law:** If P and T are constant

$$\text{rate of diffusion } r \propto \frac{1}{\sqrt{\rho}}$$

ρ is density

$$\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

- ❖ **Dalton's law:** $P = P_1 + P_2 + P_3 \dots\dots$

P = Total pressure

$P_1, P_2, P_3 \dots\dots$ = Pressure exerted by each component present in the mixture.

- ❖ **Ideal gas equation:** $PV = nRT = k_B NT$

n = number of moles

N = number of molecules

R = universal gas constant

k_B = Boltzmann's constant

- ❖ Pressure exerted by ideal gas

$$P = \frac{1}{3} \frac{mN}{V} \overline{v^2}$$

$$\overline{v^2} = \text{mean square velocity} \left\{ \overline{v^2} = \frac{v_1^2 + v_2^2 + v_3^2 + \dots\dots}{N} \right\}$$

m = mass of each molecule

$$\text{or } P = \frac{1}{3} nm \overline{v^2}$$

$$n = \text{number density i.e., } n = \frac{N}{V}$$

$$\text{❖ } V_{\text{rms}} = (\overline{v^2})^{1/2}$$

$$\text{K.E.} = \frac{3}{2} k_B T = \frac{1}{2} m \overline{v^2}$$

$$\overline{v^2} = \frac{3k_B T}{m}$$

$$\Rightarrow V_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$$

$$\text{❖ } V_{\text{av}} = \sqrt{\frac{8K_B T}{\pi m}}$$

$$\text{❖ } V_{\text{mp}} = \sqrt{\frac{2K_B T}{m}}$$

$$\text{❖ Mean free path } (\overline{l}) = \frac{1}{\sqrt{2} n \pi d^2}$$

n = number density

d = diameter of molecule

$$\text{❖ } \gamma = \frac{C_p}{C_v}$$

γ = ratio of specific heats

C_p = specific heat at constant pressure

C_v = specific heat at constant volume

$$C_p - C_v = R$$

R = universal gas constant

S.No.	Atomicity	No. of degree of freedom	C_p	C_v	$\gamma = C_p/C_v$
1	Monoatomic	3	$\frac{5}{2}R$	$\frac{3}{2}R$	$\frac{5}{3}$
2	Diatomic	5	$\frac{7}{2}R$	$\frac{5}{2}R$	$\frac{7}{5}$

$$C_{v(\text{mix})} = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}$$

where n_1 and n_2 are number of moles of gases mixed together
 C_{v_1} and C_{v_2} are molar specific heat at constant volume of the two gas and $C_{v(\text{mix})}$: Molar specific heat at constant volume for mixture.

THERMODYNAMICS

- ❖ First law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

- ❖ Work done.

$$\Delta W = P\Delta V$$

$$\therefore \Delta Q = \Delta U + P\Delta V$$

- ❖ Relation between specific heats for a gas

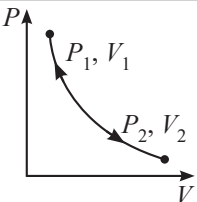
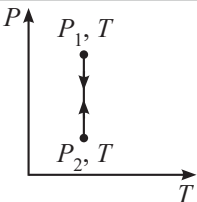
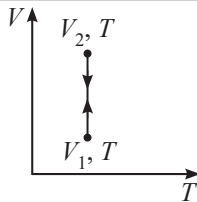
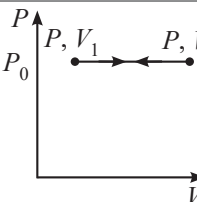
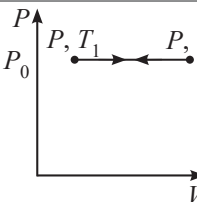
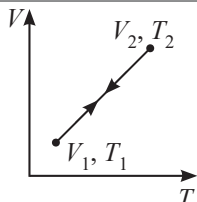
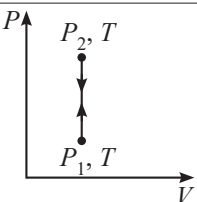
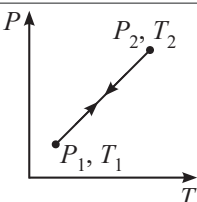
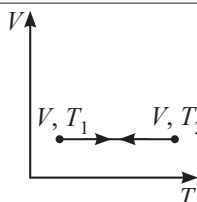
$$C_p - C_v = R$$

Polytropic Process

It is a thermodynamic process that can be expressed as follows:

$$PV^x = \text{Constant}$$

x (Polytropic exponent)	Type of standard process	Expression
0	Isobaric ($dP = 0$)	$P = \text{Constant}$
1	Isothermal ($dT = 0$)	$PV = \text{Constant}$
γ	Adiabatic ($dQ = 0$)	$PV^\gamma = \text{Constant}$
∞	Isochoric ($dV = 0$)	$V = \text{Constant}$

Process	Definition	P-V graph	P-T graph	V-T graph
Isothermal	Temperature constant			
Isobaric	Pressure constant			
Isochoric	Volume constant			

For Polytropic Process

$$\Delta Q = \Delta U + W$$

$$\Rightarrow nC\Delta T = \frac{f}{2}nR\Delta T + \frac{nR\Delta T}{1-x}$$

$$\Rightarrow C = \frac{f}{2}R + \frac{R}{1-x}$$

For infinitesimal changes in Q , U , and W , we can write,

$$dQ = dU + dW$$

$$\Rightarrow nCdT = \frac{f}{2}nRdT + PdV$$

$$\Rightarrow C = \frac{f}{2}R + \frac{P}{n} \frac{dV}{dT}$$

[$\therefore$ Work done in a general polytropic process = $[nR\Delta T/(1-x)]$]

Process	Equation of State	W	ΔU
Isobaric ($dP = 0$)	$\frac{V}{T} = c$	$P(V_f - V_i) = nR(T_f - T_i)$	$\frac{f}{2} nR(T_f - T_i) = \frac{f}{2} P(V_f - V_i)$
Isochoric ($dV = 0$)	$\frac{P}{T} = c$	0	$\frac{f}{2} nR(T_f - T_i) = \frac{f}{2} V(P_f - P_i)$
Isothermal ($dT = 0$)	$PV = c$	$nRT \ln\left(\frac{V_f}{V_i}\right) = nRT \ln\left(\frac{P_i}{P_f}\right)$	0
Adiabatic ($dQ = 0$)	$PV^\gamma = c$	$\frac{f}{2} nR(T_i - T_f) = \frac{f}{2} (P_i V_i - P_f V_f)$	$\frac{f}{2} nR(T_f - T_i)$

Process	ΔQ
Isobaric ($dP = 0$)	$\left(\frac{f}{2} + 1\right) nR\Delta T = \left(\frac{f}{2} + 1\right) P(V_f - V_i)$
Isochoric ($dV = 0$)	$\frac{f}{2} nR(T_f - T_i) = \frac{f}{2} V(P_f - P_i)$
Isothermal ($dT = 0$)	$nRT \ln\left(\frac{V_f}{V_i}\right) = nRT \ln\left(\frac{P_i}{P_f}\right)$
Adiabatic ($dQ = 0$)	0

❖ Slope of adiabatic = γ (slope of isotherm)

❖ Carnot engine

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$W = Q_1 - Q_2$$

$$(\text{efficiency}) \eta = \frac{W}{Q_1}$$

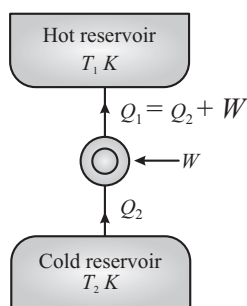
❖ Refrigerator

Coefficient of performance is β

$$\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{Q_2}{W}; \quad \beta = \frac{1 - \eta}{\eta}$$

❖ Heat pump

$$\alpha = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} = \frac{1}{\eta}$$



SUMMARY

- ❖ Zeroth law of thermodynamics states that 'two systems in thermal equilibrium with a third system are in thermal equilibrium with each other.'
- ❖ Zeroth law leads to the idea of temperature.
- ❖ Heat and work are two modes of energy transfer to the system.

- ❖ Heat gets transferred due to temperature difference between the system and its environment (surroundings).
- ❖ Work is energy transfer which arises by other means, such as moving the piston of a cylinder containing the gas, by raising or lowering some weight connected to it.
- ❖ Internal energy of any thermodynamic system depends only on its state. The internal energy change in a process depends only on the initial and final states, not on the path, i.e. it is state function.
- ❖ The internal energy of an isolated system is constant.
- ❖ (a) For isothermal process $\Delta T = 0$
(b) For adiabatic process $\Delta Q = 0$
- ❖ First law of thermodynamics states that when heat Q is added to a system while the system does work W , the internal energy U changes by an amount equal to $Q - W$. This law can also be expressed for an infinitesimal process.
- ❖ First law of thermodynamics is general law of conservation of energy.
- ❖ Second law of thermodynamics does not allow some processes which are consistent with the first law of thermodynamics. It states

Clausius statement: No process is possible whose only result is the transfer of heat from a colder object to a hotter object.

Kelvin-Planck statement: No process is possible whose only result is the absorption of heat from a reservoir and complete conversion of the heat into work.

- ❖ No engine can have efficiency equal to 1 or no refrigerator can have co-efficient of performance equal to infinity.
- ❖ Carnot engine is an ideal engine.
- ❖ The Carnot cycle consists of two reversible isothermal process and two reversible adiabatic process.
- ❖ If $Q > 0$, heat is added to the system.
If $Q < 0$, heat is removed from the system.
If $W > 0$, work is done by the system (Expansion).
If $W < 0$, work is done on the system (Compression).